TrialWatch: A Clinical Trial Compliance Intelligence System

Turning clinical trial noncompliance into visible, actionable risk.

trialwatch-flask.onrender.com

TrialWatch

CLINICAL TRIAL COMPLIANCE INTELLIGENCE

BACKGROUND

Problem & Opportunity

The Problem

Under FDAAA 801 (2007), all Applicable Clinical Trials must report results within 12 months of completion. 18 years later, over 70% of trials are noncompliant — with no single system connecting who owes results, what drugs were tested, what adverse events occurred, and how much taxpayer money is at stake.

The Opportunity

TrialWatch is a Big Data compliance intelligence platform that aggregates four federal public datasets to surface noncompliant sponsors, link adverse event reports, and quantify taxpayer funding at risk — all in one searchable, live, publicly accessible interface.

Who It's For

Regulators	Enforcement lists & sponsor compliance tracking
Journalists	Sponsor exposés, danger data, public accountability
Pharma compliance teams	Real-time deadline monitoring & risk scoring

92,757
Noncompliant Trials

$1.6B
NIH Funding at Risk

65.7M
FAERS Rows Processed

33 yrs
Worst Offender

Big Data Project
PySpark · MongoDB Atlas · Flask · ClinicalTrials.gov · FDA FAERS · NIH Reporter

Four Federal Datasets - Zero Licensing Cost

ClinicalTrials.gov

FREE

SQL-adjacent / REST API · ~500K trials

Output: Compliance status & ACT filtering

8 key fields extracted; filtered to Applicable Clinical Trials only. Primary source for trial metadata, dates, and sponsor info.

FDA FAERS

FREE

Bulk CSV + REST API · 28M+ AE reports

Output: Danger scoring & AE linkage

Adverse event reports matched to drug ingredients per trial. Used to compute danger scores and risk tiers (CRITICAL ⟶ LOW).

NIH Reporter

FREE

REST API · Federal grant data

Output: Public funding at risk per sponsor

Grant funding linked to noncompliant trials by org name. Surfaces how much taxpayer money funds unreported research.

OpenFDA Drug Labels

FREE

REST API · Drug label database

Output: Brand ↔ generic standardization

Ingredient normalization across all four datasets. Enables accurate cross-source matching of drug names and trial sponsors.

All four sources are federally mandated public datasets — zero licensing costs, zero vendor lock-in. This is a core competitive advantage.

DATA ARCHITECTURE

What We Actually Built

Step 1 — Ingestion

- Source: ClinicalTrials.gov — 581,102 trials
- Processed with PySpark on Google Colab

ACT Trial filter - 4 criteria: Interventioanl study type, FDA-regulated drug, biologic, or devic, Primary completion date after Dec 26, 2007, & Not Pase 1

- 581,102 → 132,185 applicable trials
- Compliance categories: COMPLIANT / LATE / MISSING / NOT DUE YET
- Result: 70.2% non-compliant (LATE or MISSING)

Step 2 — Enrichment

- Added a danger score and public dollars at risk to each non-compliant trial
- FDA FAERS — 65.7M rows processed with PySpark on Google Colab
- 43.6% of trials matched to a drug with adverse event data
- Danger score assigned 0–100 via log normalization of adverse event counts: CRITICAL / HIGH / MODERATE / LOW / NO DATA
- NIH Reporter API → $11.1B funding across 6,618 orgs

Step 3 — Storage & Query

- MongoDB Atlas M0 (free tier) — 3-collection schema
- trials · compliance_status · risk_enrichment
- PyMongo — loaded CSVs, parsed dates, set _id = nct_id
- $lookup aggregation joins all 3 collections by nct_id
- Indexes on org_name and overall_status
- trialwatch_queries.py — 7 reusable Python functions

Step 4 — Serving

- Flask — 6 routes (dashboard, 4 APIs, sponsor profiles)
- PyMongo — calls trialwatch_queries.py functions
- Jinja2 HTML templates + vanilla JavaScript
- Chart.js — bar charts, compliance visualisations
- Gunicorn — production WSGI server
- Render — public deployment via GitHub CI/CD

AACT flat files → PySpark (Colab) → risk_enrichment.csv → MongoDB Atlas → Flask → trialwatch-flask.onrender.com

DATA PIPELINE DIAGRAM

End-to-End Architecture

TrialWatch
CLINICAL TRIAL COMPLIANCE INTELLIGENCE

DATA SOURCES

ClinicalTrials.gov
AACT flat files · 581K trials

FDA FAERS
Bulk CSV · 65.7M AE rows

NIH Reporter
REST API · grant funding

OpenFDA Drug Labels
REST API · ingredients

INGEST

Step 1 — PySpark ingestion

581K trials → 132K ACT filter · FDAAA compliance logic · COMPLIANT / LATE / MISSING / NOT DUE YET · output: trialsclean.csv + compliancemetrics.csv

ENRICH

Step 2 — API + PySpark enrichment

FAERS matching (65.7M rows) · NIH Reporter funding lookup · danger score 0–100 via log normalization · 43.6% matched · $1.6B NIH at risk · output: risk_enrichment.csv

STORE & QUERY

Step 3 — MongoDB Atlas

3 collections: trials · compliance_status · risk_enrichment · $lookup aggregation joins · indexes on org_name + overall_status · PyMongo driver · 7 reusable query functions → trialwatch_queries.py

SERVE

Step 4 — Flask + Gunicorn + Render · trialwatch-flask.onrender.com

6 routes · Jinja2 HTML templates · vanilla JavaScript · CSS · Chart.js bar & compliance charts · CI/CD via GitHub · live public dashboard

PROS / CONS OF DATA ARCHITECTURE TOOLS

Why We Chose These Tools

MongoDB Atlas (NoSQL)

Pros

- Flexible schema for trial + enrichment data
- $lookup joins all 3 collections
- Free M0 tier — $0 for prototype

Cons

- No native joins — verbose aggregation
- 512MB free tier — M10 needed at scale

PySpark (Big Data processing)

Pros

- Handles 65.7M FAERS rows — pandas would crash
- Processes 581K trials in parallel
- Same code runs on Colab and AWS EMR

Cons

- JVM startup overhead on Colab
- Sessions disconnect — needed checkpoints

Flask (Web framework)

Pros

- Pure Python — no context switching
- Free on Render with Gunicorn
- PyMongo connects in 3 lines

Cons

- Single-threaded — needs load balancer at scale
- Not built for horizontal scaling

API (Data Fetching)

Pros

- Targeted queries — fetch only what you need, no bulk download
- Structured JSON responses — no parsing overhead
- NIH Reporter queried 6,618 orgs in minutes

Cons

- Not viable for large datasets: FAERS via API would have taken days
- Dependent on server availability

BUSINESS MODEL & REVENUE

Revenue Streams & Market Opportunity

Stream 1: SaaS Dashboard Subscriptions

Tier	$/month	Target Customer
Basic	**$500**	Researchers, small consultancies, journalism orgs
Professional	**$2,000**	Mid-size pharma, university offices, CROs
Enterprise	**$5–10K**	Large pharma, FDA, NIH, law firms, govt agencies

Stream 2: AI Chatbot Add-On (USP)

$500–$1,000/mo add-on | Natural language queries over compliance data | Mobile app for field intelligence

Stream 3: Custom Intelligence Reports

Congressional offices $5–15K · Law firms $10–25K · Journalists $2–5K
High-margin, project-based engagements for tailored compliance analysis

$2.4B
CTMS Market (2025)
15% CAGR → $4.9B by 2030

$2.7B
AI-in-Trials Market
24–28% CAGR → $8.5B by 2030

$0
Data Licensing Cost

0
Direct Competitors

Key Financial Metrics

78–85%	29:1	8–10 mo
Gross Margin	LTV:CAC	Payback

FINANCIAL FEASIBILITY & COST ANALYSIS

3-Year Projection & Infrastructure Costs

	Year 1	Year 2	Year 3
Enterprise SaaS	$480K	$1,440K	$3,600K
Professional SaaS	$360K	$840K	$1,440K
Basic SaaS	$120K	$300K	$720K
Custom Reports	$100K	$220K	$400K
Total Revenue	**$1,060K**	**$2,800K**	**$6,160K**
Total Costs	($800K)	($1,135K)	($1,730K)
Net Profit	**$260K**	**$1,665K**	**$4,430K**
Net Margin	24.5%	59.5%	71.9%

3-Year Cumulative Profit: ~$6.36M | Break-even: Month 8–10 of Year 1

Year 1 Cost Breakdown

Personnel $650K
(2 Data Eng + 1 Full-Stack + 1 Product Lead + 1 PT Data Scientist)
Infrastructure $45K · LLM API $15K · Legal $20K · Marketing $40K · Other $30K

$0/month

Current Prototype Infrastructure
Colab + Docker + Supabase Free + MongoDB Free

$2.2–$5.3K/mo

Production Deployment (Year 1)
<7% of revenue — healthy SaaS benchmark

Production Cost Breakdown

AWS EMR (Spark)	**$800–1,500**
AWS MSK (Kafka)	**$200–400**
AWS S3 (Data Lake)	**$50–100**
Supabase Pro (PostgreSQL)	**$25**
MongoDB Atlas M10	**$57**
Managed Airflow (MWAA)	**$350–400**
LLM API (Chatbot)	**$500–2,000**

SCALABILITY

Scaling Beyond the Prototype

Current Prototype — $0/month

- PySpark processing
- MongoDB Atlas - free tier
- Supabase free PostgreSQL
- Render free web service (Flask)
- Docker Compose local Kafka
- GitHub for version control

Production Scale — $2.2–5.3K/month

- AWS EMR (managed Spark) — $800–1,500/mo
- AWS MSK (managed Kafka) — $200–400/mo
- MongoDB Atlas M10 — $57/mo
- AWS S3 (data lake) — $50–100/mo
- Supabase Pro — $25/mo
- Managed Airflow (MWAA) — $350–400/mo

Scaling Challenges & Solutions

Flask cannot scale horizontally

→ Replace with FastAPI + Gunicorn + load balancer (e.g. AWS ALB)

MongoDB free tier 512MB limit

→ Atlas M10+ with sharding for distributed NoSQL at scale

Colab has session limits

→ Migrate Spark jobs to AWS EMR or Databricks managed clusters

Kafka on Docker not production-ready

→ AWS MSK (Managed Kafka) with multi-AZ replication

DATA GOVERNANCE

Security, Quality & Compliance

Application Security

- MongoDB URI stored as environment variable — never committed to Git
- .env + .gitignore enforced — credentials excluded from GitHub repo
- HTTPS enforced on Render deployment — all traffic encrypted
- No user authentication required (public data, public dashboard)

Data Architecture Security

- MongoDB Atlas network access restricted by IP allowlist
- Delta Lake ACID transactions prevent partial writes and corruption
- Great Expectations validates schema on every pipeline run
- dbt Core enforces data contract between transformation layers

Quality & Licensing

- All 4 data sources are federally mandated public datasets — no licensing
- ClinicalTrials.gov, FAERS, NIH Reporter, OpenFDA: all openly licensed
- dbt documentation + OpenMetadata catalog for data lineage tracking
- Great Expectations: schema checks, null checks, value range validation

Regulatory Awareness

- FDAAA 801: primary law we track compliance for — no PII collected
- No HIPAA concern: we process aggregate trial metadata, not patient data
- No GDPR concern: all data is federal/public; no EU personal data stored
- Open Government Data Act: supports our right to use and publish this data

DEMO

What We Built

Using Flask, MongoDB Atlas, and PySpark, we built TrialWatch — a live compliance intelligence dashboard demonstrating what we learned this semester across ingestion, storage, processing, and serving.

PySpark

Processed 581K AACT trials + 65.7M FAERS rows on Google Colab

MongoDB Atlas

NoSQL store — trials, compliance_status, risk_enrichment collections

PyMongo

Python driver — $lookup aggregation joins, indexes, query functions

Flask

6 routes: dashboard, 4 JSON APIs, sponsor profile pages

Chart.js

Compliance bar chart, overdue sponsors horizontal bar chart

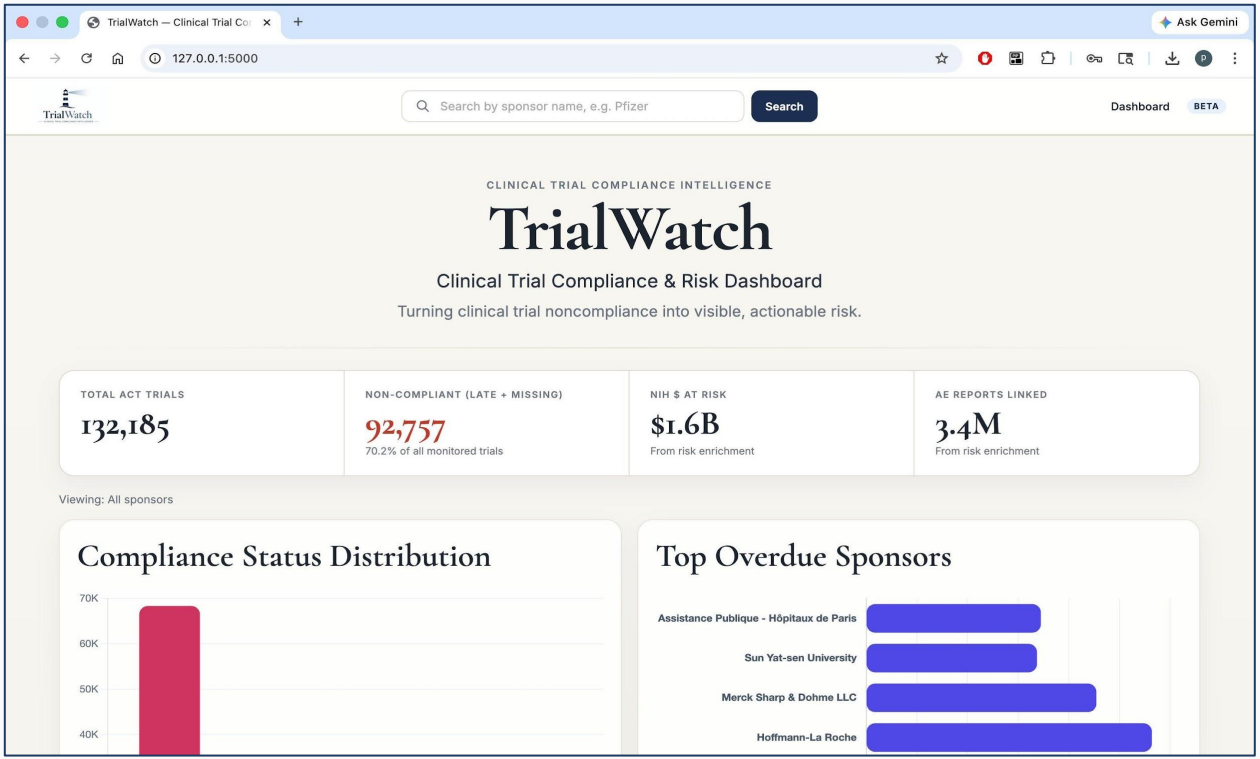
Render + GitHub

CI/CD — every git push triggers an automatic live redeploy

Live at: trialwatch-flask.onrender.com

DEMO — DASHBOARD

Live Dashboard — KPIs & Compliance Charts



132,185 Total ACT Trials monitored

92,757 Noncompliant (70.2%)

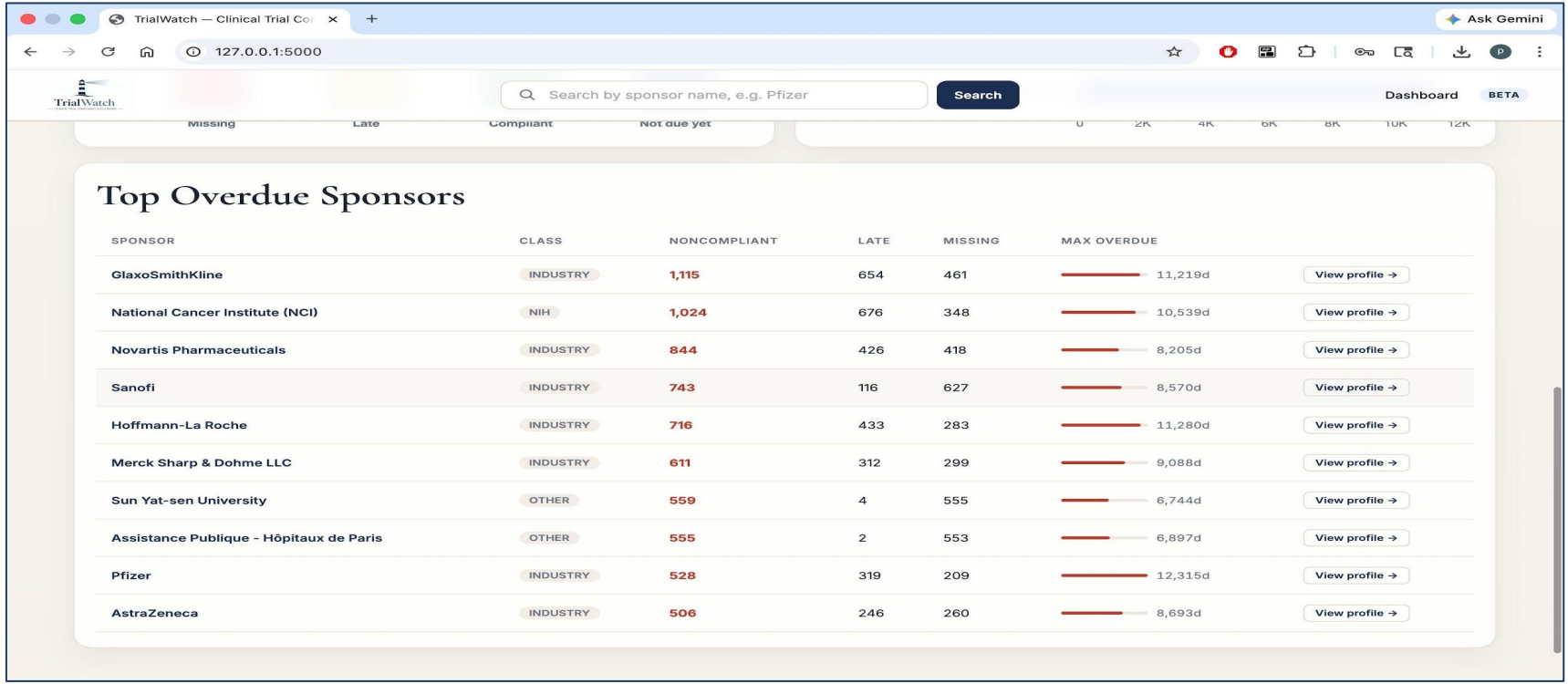
$1.6B NIH funding at risk

3.4M From risk enrichment

Real-time data from MongoDB Atlas — updated on every page load.

DEMO — TOP OVERDUE SPONSORS

Sponsor Leaderboard — Who Owes Results?



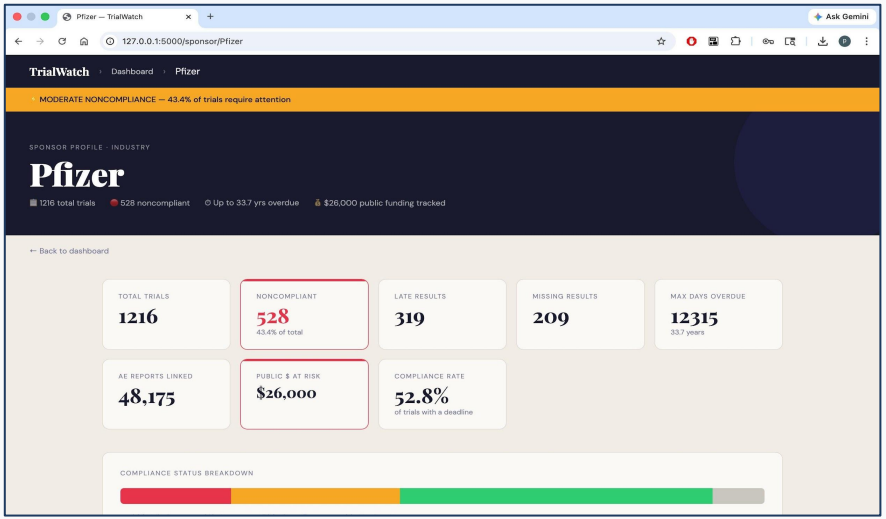
Top Overdue Sponsors

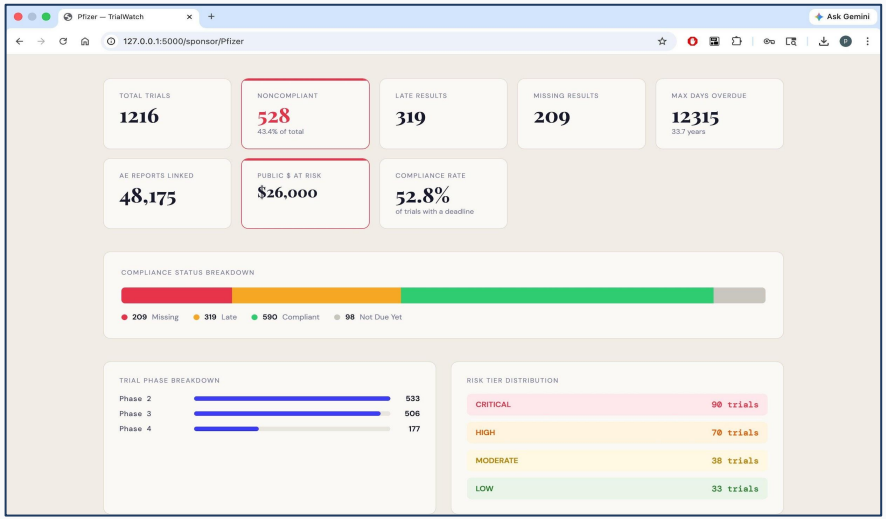
SPONSOR	CLASS	NONCOMPLIANT	LATE	MISSING	MAX OVERDUE	
GlaxoSmithKline	INDUSTRY	1,115	654	461	11,219d	View profile →
National Cancer Institute (NCI)	NIH	1,024	676	348	10,539d	View profile →
Novartis Pharmaceuticals	INDUSTRY	844	426	418	8,205d	View profile →
Sanofi	INDUSTRY	743	116	627	8,570d	View profile →
Hoffmann-La Roche	INDUSTRY	716	433	283	11,280d	View profile →
Merck Sharp & Dohme LLC	INDUSTRY	611	312	299	9,088d	View profile →
Sun Yat-sen University	OTHER	559	4	555	6,744d	View profile →
Assistance Publique - Hôpitaux de Paris	OTHER	555	2	553	6,897d	View profile →
Pfizer	INDUSTRY	528	319	209	12,315d	View profile →
AstraZeneca	INDUSTRY	506	246	260	8,693d	View profile →

Every sponsor name is clickable — links to a full intelligence profile page. Mini bar charts show relative overdue severity at a glance.

DEMO — SPONSOR INTELLIGENCE PROFILE

Pfizer Profile — Full Sponsor Intelligence





Risk Banner: Auto-classifies HIGH / MODERATE / LOW based on noncompliance %

8 Stat Cards: Total, noncompliant, late, missing, AEs, dollars, max overdue, rate

Phase Breakdown: Trials grouped by Phase 1–4 with proportional bar charts

Risk Tier Dist.: CRITICAL / HIGH / MODERATE / LOW tiers from FAERS danger scores

DEMO — TRIAL-LEVEL DETAIL

Most Overdue Trials — NCT ID Level

Pfizer — TrialWatch

127.0.0.1:5000/sponsor/Pfizer

Most Overdue Trials

NCT ID	STATUS	COMPLIANCE	DAYS OVERDUE	AE REPORTS	PHASES	PRIMARY COMPLETION
NCT00001017	Completed	MISSING	12315	–	Ph3	1991-07-31
NCT00751127	Completed	MISSING	11372	–	Ph3	1994-02-28
NCT00003673	Completed	MISSING	9515	–	Ph2	1999-03-31
NCT00003343	Completed	MISSING	8843	–	Ph3	2001-01-31
NCT00003293	Completed	MISSING	8723	–	Ph3	2001-05-31
NCT00004199	Completed	MISSING	8662	–	Ph3	2001-07-31
NCT00527046	Terminated	MISSING	8570	–	Ph2	2001-10-31
NCT00004200	Completed	MISSING	8478	–	Ph2	2002-01-31
NCT00003131	Completed	MISSING	8450	–	Ph2	2002-02-28
NCT00034957	Completed	MISSING	8205	–	Ph2	2002-10-31

TrialWatch · Clinical Trial Compliance Intelligence · Data sourced from ClinicalTrials.gov ACT dataset

12,315

days overdue
(NCT00001017)

33.7 yrs

single trial
never reported

Every NCT ID links directly to ClinicalTrials.gov

All 10 worst trials shown — each NCT ID is a live hyperlink to the official federal database entry.

Q&A

Thank You

Each team member contributed to ingestion, processing, storage, and the live dashboard deployment.

Github:
https://github.com/prineet3/TrialWatch

TrialWatch

trialwatch-flask.onrender.com

132K	92,757	$1.6B	$0
Trials	Violations	At Risk	Data Cost

PySpark · ClinicalTrials.gov API · FDA FAERS · NIH Reporter · MongoDB Atlas · PyMongo · Flask · Chart.js · Render